Probability Theory and Statistics
Lecture 15

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Conditional Expectation: discrete case

We now formulate the example and the definitions in the general discrete setting.

Definition

Let Y be a random variable and A an event with $\mathbb{P}(A) > 0$. The conditional expectation of Y given A is the expectation of Y with respect to the probability measure $\mathbb{P}(\cdot|A)$. Notation: $\mathbb{E}(Y|A)$.

Lemma

Let Y be a simple random variable and A an event with $\mathbb{P}(A) > 0$. Then

$$\mathbb{E}(Y|A) = \sum_{k \in \mathrm{Ran}(Y)} k\mathbb{P}(Y = k|A).$$

Regression in the continuous case: first example

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Proper answer: since $\mathbb{P}(X > 1/2) = \frac{1}{2} > 0$, by the previous definition

$$\mathbb{E}(X|X > 1/2) = \frac{\mathbb{E}\big(X \cdot \mathbb{1}_{\{X \in (1/2,1)\}}\big)}{\mathbb{P}(X > 1/2)} = \frac{\int_{1/2}^{1} x f_X(x)\,\mathrm{d}x}{1/2}$$

$$= \frac{\int_{1/2}^{1} x\,\mathrm{d}x}{1/2} = \frac{1/2 - 1/8}{1/2} = \frac{3}{4}.$$

Regression in the continuous case

Assume now that X is continuous. For a given Y we want to define $\mathbb{E}(Y|X)$ as the „best approximation to Y given X," analogous to the discrete case.

For a discrete Z we had:

$$\mathbb{E}[Y|Z=z] = \sum_{k \in \mathrm{Ran}(Y)} k\mathbb{P}(Y=k|Z=z),$$

provided $\mathbb{P}(Z=z) > 0$.

Problem: for continuous X we have $\mathbb{P}(X=x) = 0$ for all $x \in \mathbb{R}$. Hence the formula above and classical conditional probabilities do not define $\mathbb{E}(Y|X=x)$!

This motivated the conditional density when (X, Y) is a continuous random vector:

$$f_{Y|X}(y|x) = \frac{f_{(X,Y)}(x,y)}{f_X(x)},$$

if $f_X(x) > 0$, and 0 otherwise.

Conditional density and regression

Conditional density:

$$f_{Y|X}(y|x) = \frac{f_{(X,Y)}(x,y)}{f_X(x)},$$

if $f_X(x) > 0$, and 0 otherwise.

Earlier (informally) we said this behaves like $f_Y(y)$, but conditional on X being near x.

We know $\mathbb{E}(Y) = \int_{-\infty}^{\infty} y f_Y(y)\,\mathrm{d}y$. Thus we expect that

$$\mathbb{E}(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)\,\mathrm{d}y$$

is a good definition. (This is again formal notation, not classical conditional probability.)

Introduce $g(x) = \mathbb{E}(Y|X = x)$. In the discrete case we had $\mathbb{E}(Y|X) = g(X)$.

Question: Is the same true more generally (e.g., for continuous X)? What is the definition of regression in that case?

Example

On a poorly maintained railway branch line, the waiting time for the train is $X \sim \mathrm{Exp}(1/10)$ minutes, and given $X = x$, the travel time to the destination is uniformly distributed on the interval $(x, 2x)$, i.e. $Y \sim \mathrm{U}(x, 2x)$.

- Determine $f_{Y|X}$ and $f_{X,Y}$.
- Determine the regression function $g(x) = \mathbb{E}(Y|X = x)$ of Y on X.

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Since $X \sim \mathrm{Exp}(1/10)$,

$$f_X(x) = \frac{1}{10}\,\mathrm{e}^{-x/10},$$

for $x > 0$, and $f_X(x) = 0$ otherwise. Furthermore, given $X = x$, we have $Y \sim \mathrm{U}(x, 2x)$, hence the conditional density is

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{1}{2x - x} = \dfrac{1}{x}, & \text{if } x < y < 2x, \\ 0, & \text{otherwise.} \end{cases}$$

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- Determine $f_{Y|X}$ and $f_{X,Y}$.
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Therefore, using $f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y|x)$ (for $f_X(x) > 0$),

$$f_{X,Y}(x, y) = \begin{cases} \dfrac{1}{10x}\,\mathrm{e}^{-x/10}, & \text{if } 0 < x < y < 2x, \\ 0, & \text{otherwise.} \end{cases}$$

Law of Total Expectation

Example: How can we compute the expected value of Y in the train example?
Of course, one could compute it from $f_{X,Y}$ or f_Y in the usual way.
A simpler and more elegant approach uses the following theorem.

Theorem (Law of Total Expectation)

Let X and Y be random variables such that $\mathbb{E}(|X|)$ and $\mathbb{E}(|Y|)$ are finite. Then

$$\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}(Y|X)).$$

This statement is also known as the tower property (in many languages the metaphor of a "tower" is used, e.g. German: Turmeigenschaft).

In the example: $\mathbb{E}(Y|X) = \frac{3}{2}X$, and since $X \sim \mathrm{Exp}(1/10)$, we get $\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}(Y|X)) = \frac{3}{2} \cdot 10 = 15$ minutes.

Law of Total Expectation: Discrete Case

$$\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}(Y|X)).$$

Question: how is this related to the Law of Total Probability (LTP)?

Law of Total Expectation: Discrete Case

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In the discrete case, by expanding the outer random variable, we obtain a form analogous to the LTP:

Theorem (Law of Total Expectation, Discrete Case)

Let X be a discrete random variable with possible values $\{x_1, \ldots, x_n, \ldots\}$ such that $\mathbb{P}(X = x_i) > 0$. If $\mathbb{E}(|Y|) < \infty$, then

$$\mathbb{E}(Y) = \sum_i \mathbb{E}(Y|X = x_i)\mathbb{P}(X = x_i).$$

Law of Total Expectation: Discrete Case

For a partition of the sample space, the statement looks even more like the LTP:

Theorem (Law of Total Expectation with a Partition)

Let $A_1, \ldots, A_n$ be a partition of Ω such that $\mathbb{P}(A_i) > 0$ for all i. If $\mathbb{E}(|Y|) < \infty$, then

$$\mathbb{E}(Y) = \sum_{i=1}^{n} \mathbb{E}(Y|A_i)\mathbb{P}(A_i).$$

Law of Total Expectation: Discrete and Continuous Cases

Discrete case:

$$\mathbb{E}(Y) = \sum_i \mathbb{E}(Y|X = x_i)\mathbb{P}(X = x_i).$$

(Example: the variance of the geometric distribution, Example 11.3.6.)

In the continuous case, the probability mass function $p_X(x_i) = \mathbb{P}(X = x_i)$ is replaced by the density $f_X(x)$, and the conditional expectation $\mathbb{E}(Y|X = x_i)$.

Theorem (Law of Total Expectation, Continuous Case)

Let X be a continuous random variable with $\mathbb{E}(|Y|) < \infty$, and let f_X denote its density function. Then

$$\mathbb{E}(Y) = \int_{-\infty}^{\infty} \mathbb{E}(Y|X = x) f_X(x) \mathrm{d}x.$$

Law of Total Probability for the Continuous Case

So far, we have stated the LTP for a partition of Ω. It can also be expressed for a discrete random variable: let X be discrete with $S_X = \{k \in \mathbb{R} \colon \mathbb{P}(X = k) > 0\}$. Then for any event A,

$$\mathbb{P}(A) = \sum_{k \in S_X} \mathbb{P}(X = k)\mathbb{P}(A|X = k).$$

This is a special case of the general LTP (which works equally well for countably many events).

We now wish to generalize the LTP to continuous X.

Recall that $\mathbb{P}(A) = \mathbb{E}(\mathbb{1}_A)$ for any event A.
For the continuous case, what is $\mathbb{P}(A|X = x)$ (formally)?

Definition

Let X be a random variable and A an event. The conditional probability of A given X is the regression function $x \mapsto \mathbb{E}(\mathbb{1}_A|X = x)$, denoted by $\mathbb{P}(A|X = x)$.

Law of Total Probability for the Continuous Case

For discrete X:

$$\mathbb{P}(A) = \sum_{k \in S_X} \mathbb{P}(X = k)\mathbb{P}(A|X = k).$$

The continuous analogue replaces the mass function by the density, the conditional probability by its (formal) continuous version, and the sum by an integral:

Theorem (Law of Total Probability for the Continuous Case, or "Continuous LTP")

Let X be a continuous random variable and A an event. Then

$$\mathbb{P}(A) = \int_{-\infty}^{\infty} \mathbb{P}(A|X = x) f_X(x) \mathrm{d}x,$$

where f_X is the density of X.

Proof (included here, not in the notes)

Let $g(x) = \mathbb{P}(A|X = x) = \mathbb{E}(\mathbb{1}_A|X = x)$ be the regression function. By definition, $g(X) = \mathbb{E}(\mathbb{1}_A|X)$.
By the formula for the expectation of a transformed variable,

$$\int_{-\infty}^{\infty} \mathbb{P}(A|X = x) f_X(x) \mathrm{d}x = \int_{-\infty}^{\infty} g(x) f_X(x) \mathrm{d}x = \mathbb{E}(g(X)) = \mathbb{E}(\mathbb{E}(\mathbb{1}_A|X)).$$

Finally, by the law of total expectation,

$$\mathbb{E}(\mathbb{E}(\mathbb{1}_A|X)) = \mathbb{E}(\mathbb{1}_A) = \mathbb{P}(A),$$

which completes the proof.

Law of Total Probability for the Continuous Case: Example

We have not yet discussed how to compute $\mathbb{P}(A|X = x)$ in general. .

Example 1.: Choose a λ number uniformly between 1 and 3. Let $Y \sim \mathrm{Exp}(\lambda)$. What is the probability that $Y < \frac{1}{2}$?

Law of Total Probability for the Continuous Case: Example

We have not yet discussed how to compute $\mathbb{P}(A|X=x)$ in general. .

Example 1.: Choose a λ number uniformly between 1 and 3. Let $Y \sim \mathrm{Exp}(\lambda)$. What is the probability that $Y < \frac{1}{2}$?

Example 2.: If X is a random variable with density function

$$f_X(x) = \begin{cases} 3x^2 \text{ , if } 0 < x < 1 \\ 0 \text{ , otherwise} \end{cases}$$

and $Y \sim \mathrm{Geo}(p)$, then what is the probability that $Y = 3$?